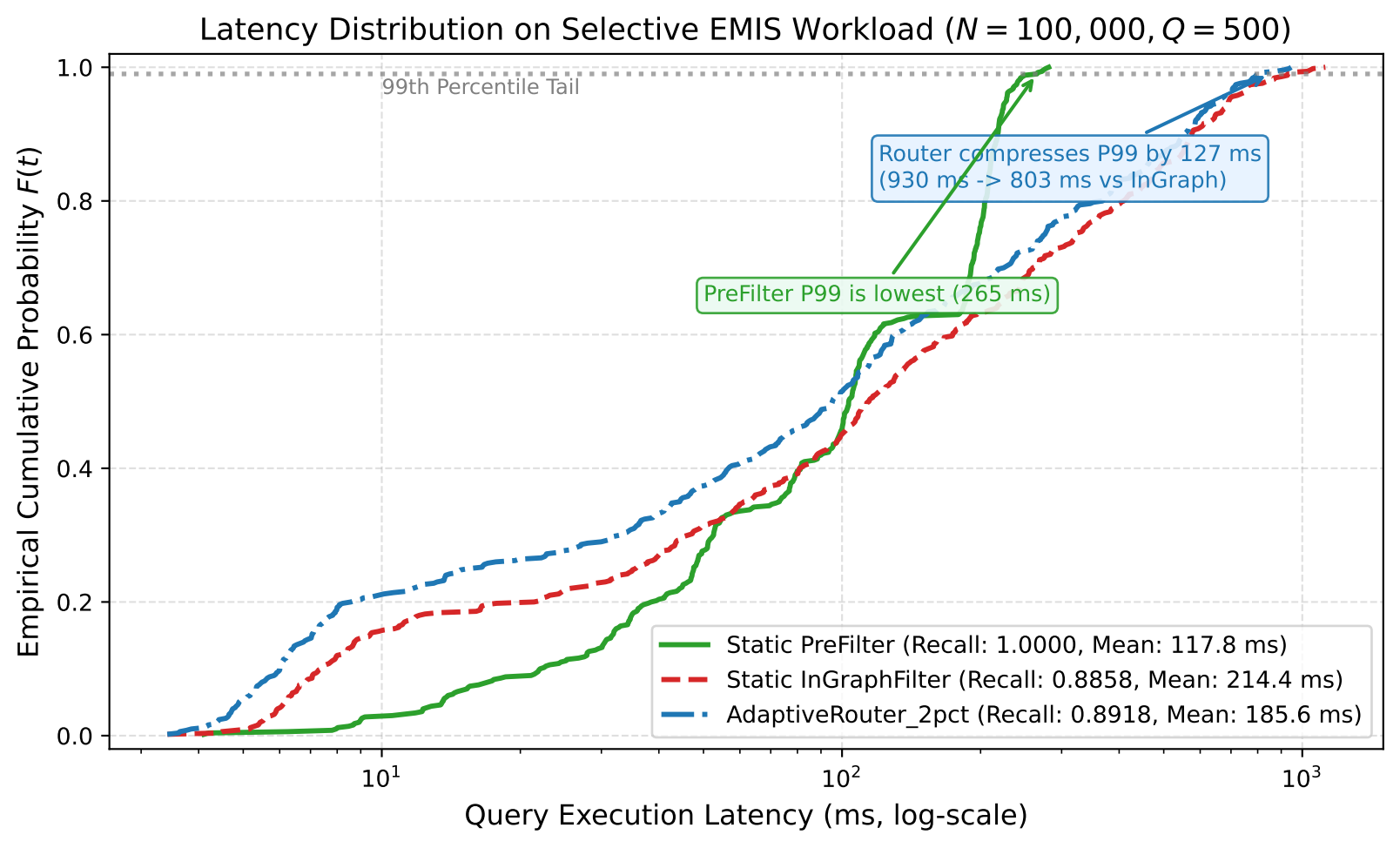
Latency Distribution on Selective EMIS Workload ($N = 100,000, Q = 500$)
99th Percentile Tail
Router compresses P99 by 127 ms
(930 ms -> 803 ms vs InGraph)
PreFilter P99 is lowest (265 ms)
Static PreFilter (Recall: 1.0000, Mean: 117.8 ms)
Static InGraphFilter (Recall: 0.8858, Mean: 214.4 ms)
AdaptiveRouter_2pct (Recall: 0.8918, Mean: 185.6 ms)
Empirical Cumulative Probability $F(t)$
Query Execution Latency (ms, log-scale)
1.0
0.8
0.6
0.4
0.2
0.0
10^1
10^2
10^3